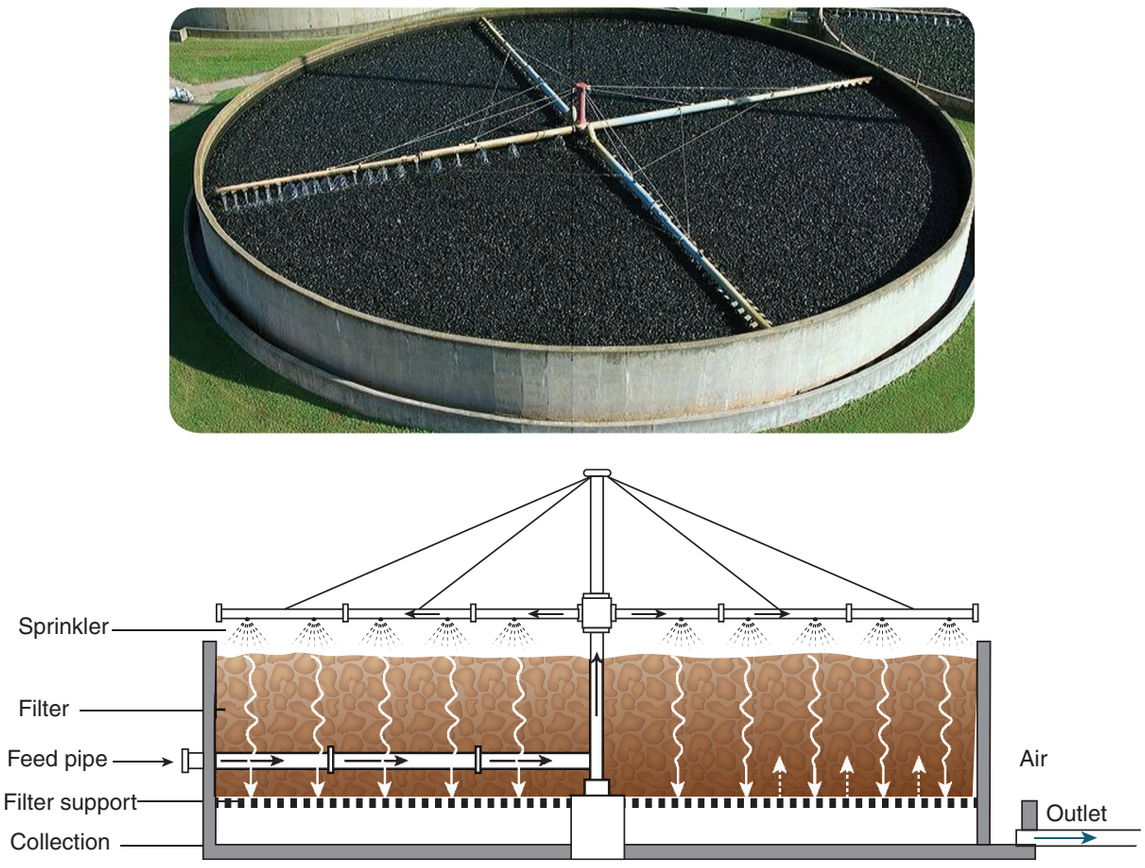


Figure 9.9: Trickling Filter in sewage treatment

solids are eliminated in sewage through filtration or sedimentation. Poorly biodegradable substances can be removed by the use of specialized microorganisms capable of using them as substrates. Chlorine is frequently added to tertiary treated effluent to kill any remaining microorganism.

9.6 Recycling of Treated Sewage

Water recycling is reusing treated wastewater for beneficial purposes such as agricultural and landscape irrigation, industrial processes, toilet flushing, and replenishing a ground water basin. Recycled water can satisfy most water demands, as long as it is adequately treated to ensure water quality appropriate for the use.

The residue that accumulates in sewage treatment plants is called sludge.

Sludge Digestion

Treatment of sewage sludge may include a combination of thickening, digestion, and dewatering processes. Among these digestion is mediated by microbes. Sludge digestion is a biological process in which organic solids are decomposed into stable substances. Digestion reduces the total mass of solids, destroys pathogens, and makes it easier to dewater or dry the sludge. Digested sludge is inoffensive, having the appearance and characteristics of a rich potting soil.

Most large sewage treatment plants use a two-stage digestion system in which organics are metabolized by bacteria anaerobically.

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